

Shea Cardozo

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RESEARCH INTERESTS

I am broadly interested in creating embodied autonomous agents and systems. My PhD work focuses chiefly on motion prediction and planning tasks for self-driving vehicles, and especially how to tackle the problems of covariate shift and causal confusion that hamper closed-loop performance of popular methods.

My current research direction is in hybrid rule-based and deep learning planners and especially the incorporation of code-generation models in human-in-the-loop rule engines. Other areas of interest of mine not directly related to my current work are model-based reinforcement learning, causal representation learning, and mechanistic interpretability techniques.

EDUCATION

2023 - 2027	PhD in Computer Science Waterloo Intelligence Systems Lab Supervisor: Professor Krzysztof Czarnecki	University of Waterloo
2021 - 2022	Master of Science in Applied Computing Data Science Concentration	University of Toronto
2017 - 2021	Bachelor of Mathematics in Statistics Co-operative Program. Dean's Honours.	University of Waterloo

SKILLS

Programming Languages	Proficiency with Python, R, and C++. Experience with Java, C#, DART, Haskell, Scheme (Racket), SQL, VBA, HTML/CSS.
Deep Learning Frameworks	Proficiency with PyTorch. Experience with Jax, TensorFlow
Development and Deployment	Git, Linux, AWS, GCP, VSCode, RStudio, Jupyter Notebooks

PUBLICATIONS

- **Shea Cardozo**, Gabriel Islas Montero, et al. 2022. *Explainer Divergence Scores (EDS): Some Post-Hoc Explanations May be Effective for Detecting Unknown Spurious Correlations*. Presented at the AIMLAI workshop at CIKM 2022. Available at [link](#).
- Eli-Henry Dykhne, **Shea Cardozo**, James Riddell, Sean Sedwards, Krzysztof Czarnecki. 2024. *Systematic Generation of Dynamic Occlusion-Caused Collisions*. Manuscript; work extended in E.-H. Dykhne, M.Sc. thesis, University of Waterloo, 2024. Available at [link](#).

AWARDS

- **University of Waterloo Graduate Scholarship** - \$5,000/year for one year Fall 2023
- **David R. Cheriton Graduate Scholarship** - \$10,000/year for two years Winter 2024

INDUSTRY EXPERIENCE

Machine Learning Scientist

May 2022 - Sept. 2023

Tenyks

- Chief Machine Learning Scientist at Tenyks (YC'S21) reporting directly to the CTO, researching and prototyping new methods for classification, detection, and segmentation tasks. Projects include:
 - Created a synthetic data pipeline for zero-shot anomaly detection via inserting photorealistic synthetic anomalies into clean data using RGB pixel masks and neural image inpainting.
 - Implemented a scalable system for open-set object retrieval based on visual and text prompting using 'Segment Anything' proposals and 'OpenCLIP' data embeddings. Fine-tuned 'OpenCLIP' embeddings on domain-specific data to improve retrieval performance beyond baseline.
 - Prototyped multiple active learning and core set approaches to suggest an efficient data annotation strategy for settings with minimal annotated data and expensive annotation costs.
 - Implemented a sequence of automated 'data quality checks' to detect common dataset issues in computer vision such as a high prevalence of occlusion or inconsistent class definitions.
 - Formulated a novel evaluation criteria for post-hoc neural network explanations to detect dependence on spurious correlations. Verified and submitted our work as a workshop paper.

Data Scientist - Claims AI Team

Sept. 2020 - Dec. 2020

Intact Insurance

- Constructed a pipeline to automatically classify insurance documents from image and text data using an ensembled 'ResNet' convolutional neural network and 'BERT' transformer neural network.
- Experimented with multi-objective non-gradient optimization methods such as the 'NSGA-II' genetic algorithm to optimize model prediction thresholds to mark unclassified documents for manual review.

Data Scientist - Analytics

May 2020 - Aug. 2020

Noom Inc.

- Specified and fit an autoregressive time series model with seasonal effects to predict the influx of user support tickets to ensure sufficient resource availability.
- Trained and benchmarked a set of 'GloVe' vector embeddings constructed from internal food data to improve user meal recommendation and tracking.

Actuarial Analyst - DataLab Division

Sept. 2019 - Dec. 2019

Intact Insurance

- As part of the 'Rating Revolution' team, trained 'XGBoost' gradient-boosted decision tree models to replace the existing generalized linear models used in home insurance pricing
- Created a Python visualization tool to analyze how different pricing models impact wider financials.

Associate Actuarial Programmer

Jan. 2019 - Apr. 2019,

Moodys Analytics

May 2018 - Aug. 2018

- Implemented highly performance sensitive financial calculations into our insurance software platform using the C++ programming language, with focus on long-term maintainability.
- Expanded UI functionality to more transparently display to clients how financials are calculated.